

Methodology:

1. Data Preparation:

- Created synthetic marketing dataset with channels, spend, impressions, clicks, conversions, and revenue.

2. KPI Calculation:

- $CTR = Clicks / Impressions$
- $CPC = Spend / Clicks$
- $CAC = Spend / Conversions$
- $ROAS = Revenue / Spend$

3. Analysis Approach:

- Channel-wise aggregation using SUMIF
- Performance comparison across channels
- Visualization using bar charts

4. Attribution Logic:

- Basic performance-based attribution assumed for channel evaluation.

5. Limitations:

- The dataset is synthetic and simplified for academic demonstration.